# Datacentric Automated Lesion Segmentation in Whole-Body PET/CT - Multitracer Multicenter Generalization

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## 1 INTRODUCTION

- This project participates in the AutoPET-III Challenge [1]
- Deep Generative Models has recently been used to synthesized medical image as a means of data augmentation for training more robust segmentation models
- We try to adapt the strategy to the PET-CT modality by generating tumours onto CT and PET images given a segmentation mask
- Results show that data augmentation enhances segmentation performance.

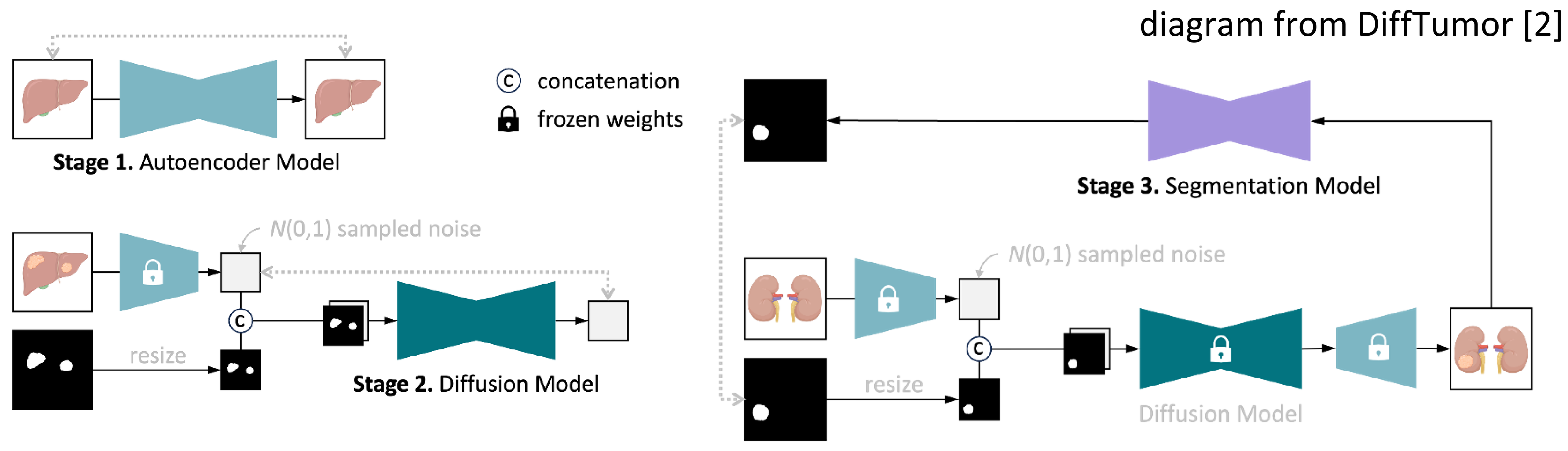
### HIghlights and Contributions

- First ever use of deep generative models for data augmentation in the task of PET-CT lesion segmentation
- Along with other similar works, collectively demonstrates the general usefulness of tumour synthesis as a means of data augmentation in segmentation tasks
- Paradigm shift in the field by focusing more on the data instead of the models to improve performance
- Significant improvement in PET-CT lesion segmentation performance using our method

## 2 METHODS

- We adapted a generative model (DiffTumor) [2] that generate tumored PET-CT image conditioned by the original healthy PET-CT image and a tumour mask
- We first train an autoencoder to learn the latent representation of PET-CT images so as to conduct the generative process in a compressed latent space.
- Then we train a latent diffusion model on the latent space under the condition of the healthy PET-CT image (encoded into latent space) as well as a tumour mask (downsized along with a organ mask)
- In the last step, we generate new synthesized tumours on the PET-CT images of the original dataset for data augmentation..
- After that, we uses the augmented dataset to train DynUnet [3, 4], the PET-CT lesion segmentation model.

diagram from DiffTumor [2]

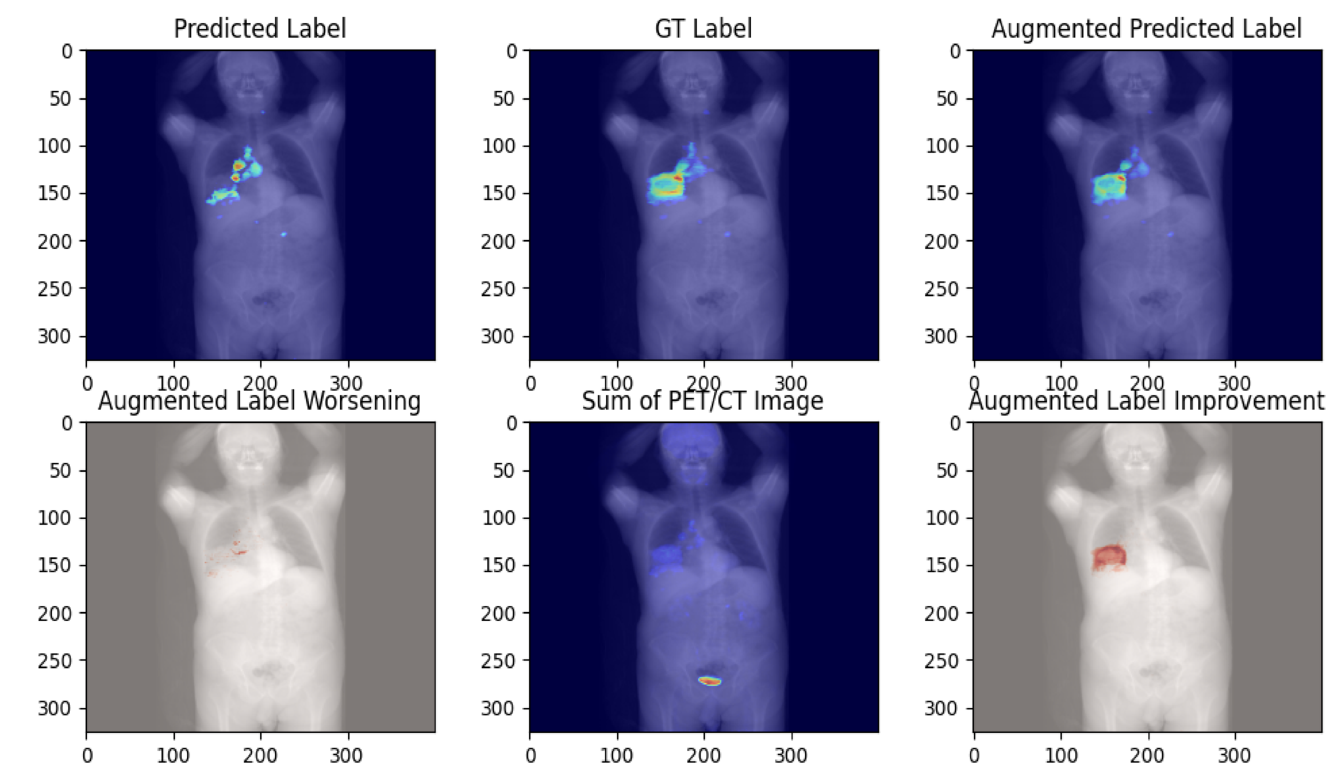


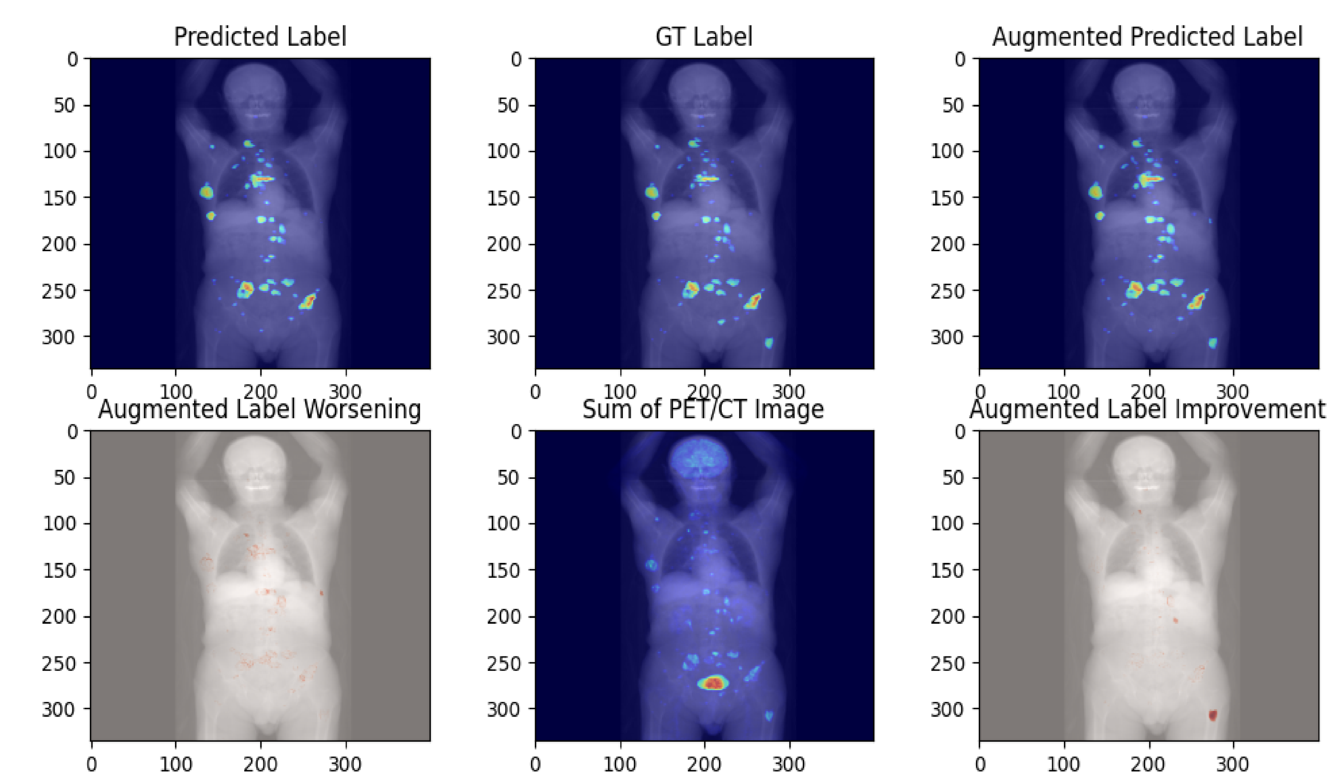

## 3 RESULTS

- shows dice scores of baseline vs augmented
- with different number of random transforms per image → different size of dataset

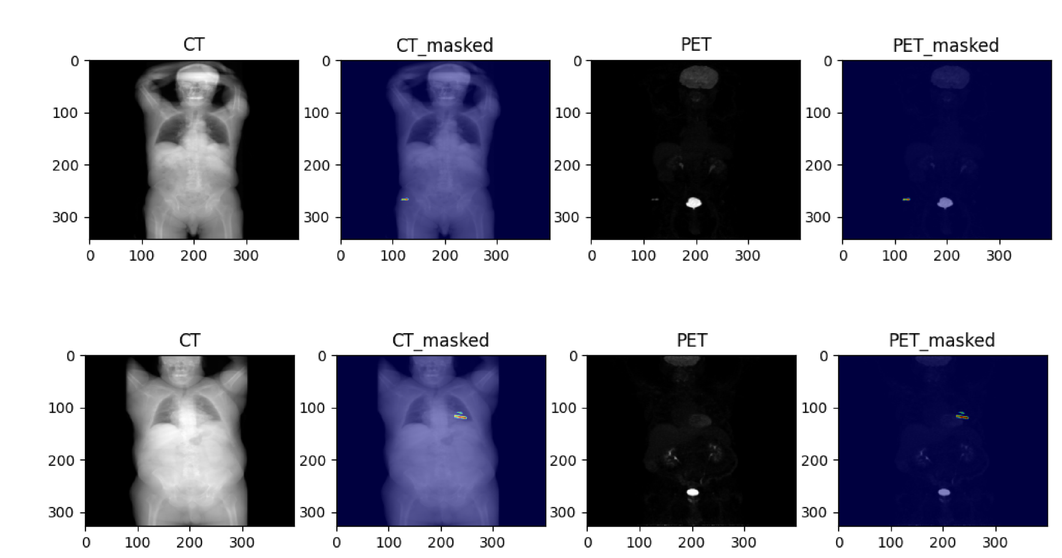
| Baseline/Augmented | Number of transforms per image | Training Dataset size | Dice Score |
|---|---|---|---|
| Baseline | 1 | 1291 | 0.3650 |
| Augmented | 1 | 5152 | 0.4542 |
| Baseline | 15 | 19365 | 0.5398 |
| Augmented | 15 | 77280 | 0.6143 |

### Lesion Segmentation Results




### Tumour Generation Results




## 4 CONCLUSION

In conclusion, we demonstrated the feasibility of this new paradigm of data augmentation through using Deep Generative Model to learn the distribution of the existing data and then synthesize data to augment the original dataset for downstream application in the task of multitracer multicentre full-body PET-CT lesion segmentation. However, further analysis is to be carried out to fully understand its benefits and downfalls, as well as the extent it can push forward the state-of-the-art in this task.

## REFERENCES

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